

New algorithms for time series classification and extrinsic regression and their application in health

Student: Isabela Guarnier De Mitri (isademitri@usp.br)

Advisor: Diego Furtado Silva (diegofsilva@icmc.usp.br)

University of São Paulo

Motivation and Objectives

Time series are present in many domains, such as finance, climatology, and health, where it is essential to balance computational efficiency and predictive performance. This study investigates time series classification algorithms [1], seeking to reduce execution cost without significant loss of accuracy. The focus is on the relationship between cost, number of kernels, and parameter values in state-of-the-art algorithms.

Methods and Procedures

DrRocket was developed as a modification of the DrCIF classifier [2], in which the original features were replaced by attributes extracted with random convolutional kernels, as proposed in the ROCKET algorithm [3].

Four configurations were evaluated: original DrCIF, DrRocket (4 kernels), DrRocket (50 kernels), and DrRocket (100 kernels). The experiments were conducted on 21 datasets from the UEA & UCR Time Series Classification Archive. The metrics used were: accuracy, F1-score, recall, precision, and execution time (training and testing).

Results and Applications

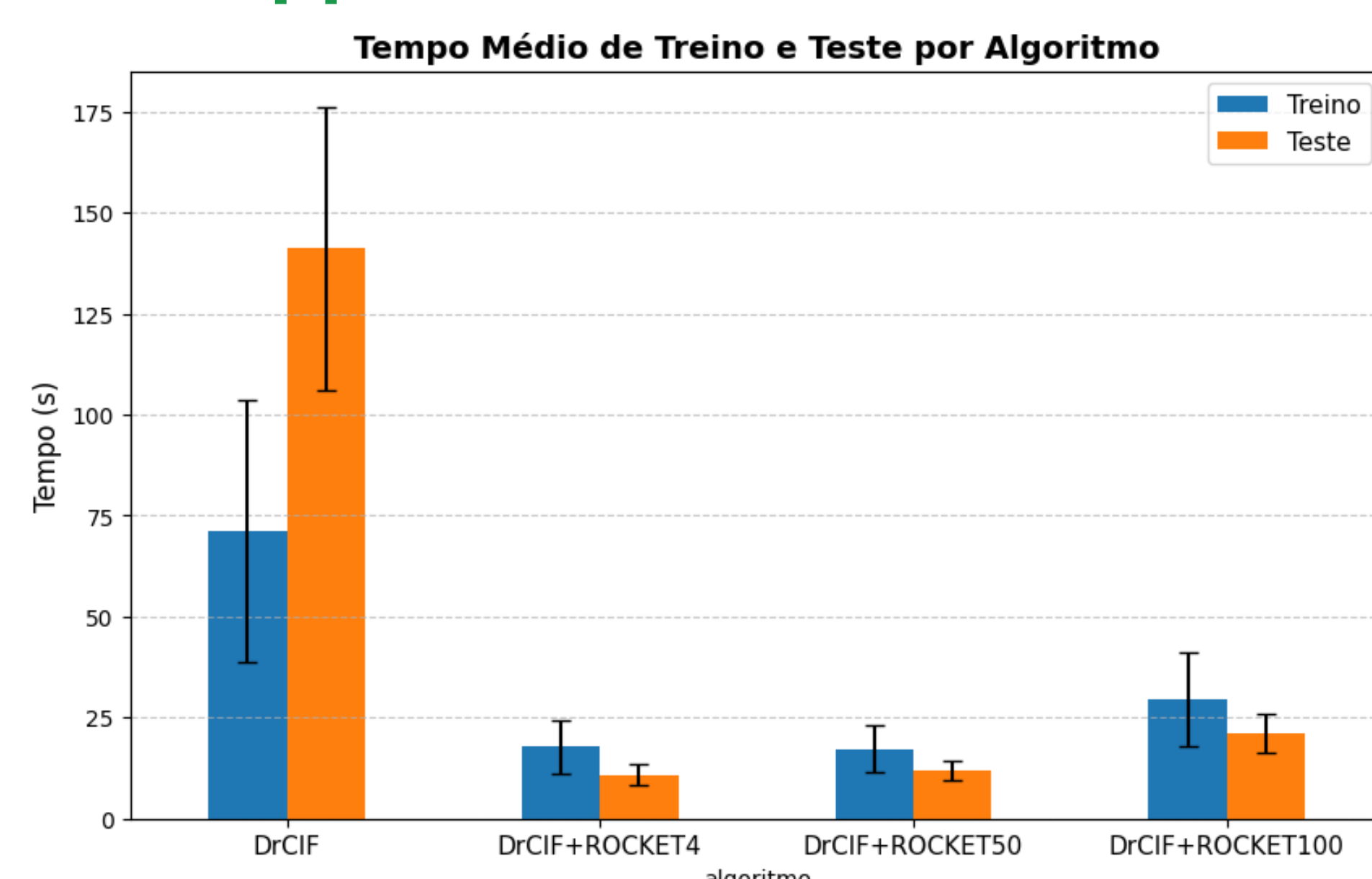


Figure 1: Average training and testing time per algorithm.

The original DrCIF showed the highest computational cost. The ROCKET-based variants significantly reduced execution time, showing feasibility in practical scenarios.

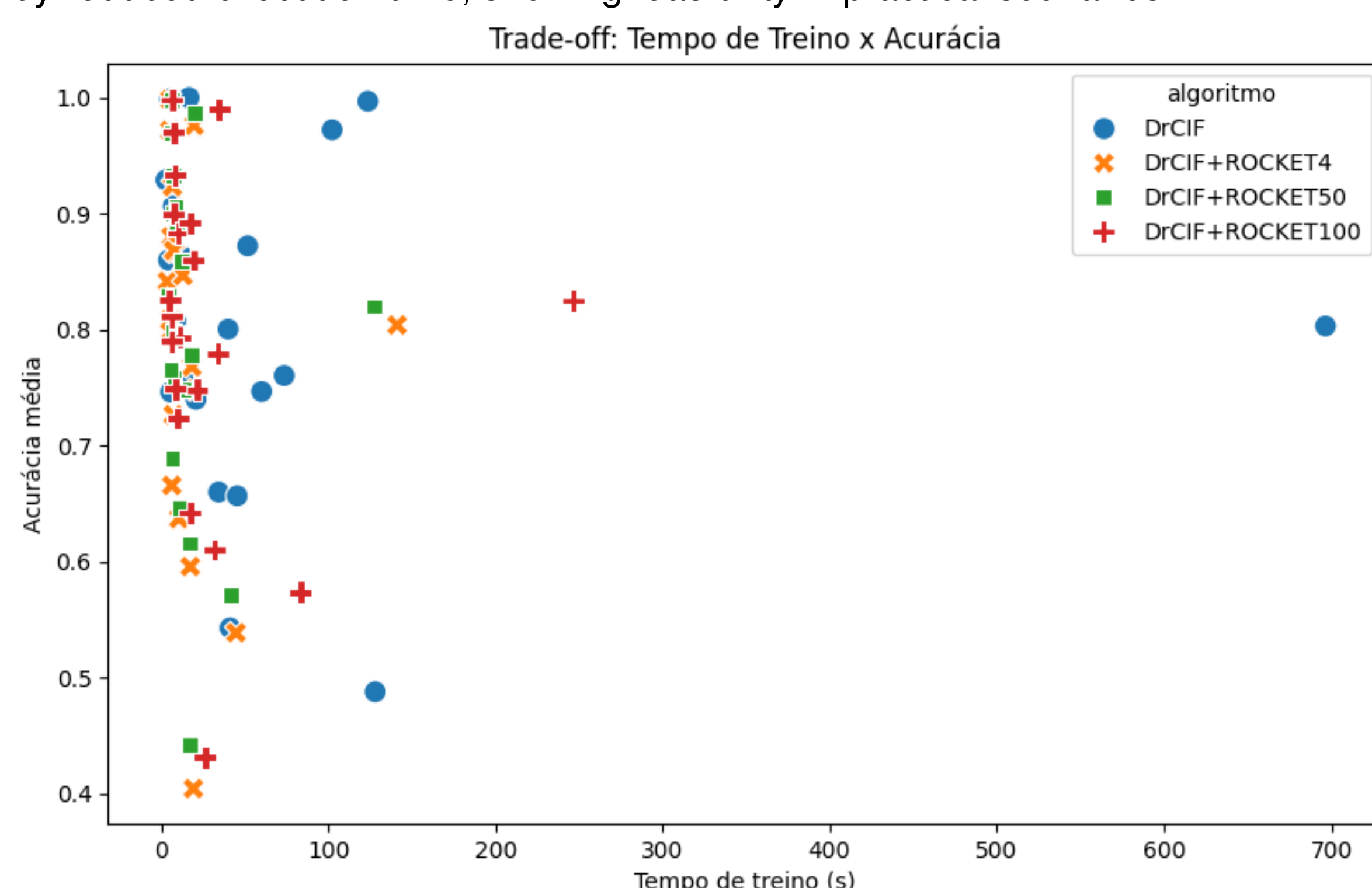


Figure 2: Trade-off between training time and accuracy.

We observe that:

- DrRocket with 4 kernels was the fastest, but showed a significant drop in accuracy;
- DrRocket with 100 kernels remained computationally expensive and did not outperform DrCIF;
- DrRocket with 50 kernels emerges as the balance point: it achieves higher accuracy than the 4-kernel model and much shorter execution time than DrCIF, proving to be the most advantageous alternative in terms of efficiency.

Diagrama de Diferença Crítica - acurácia

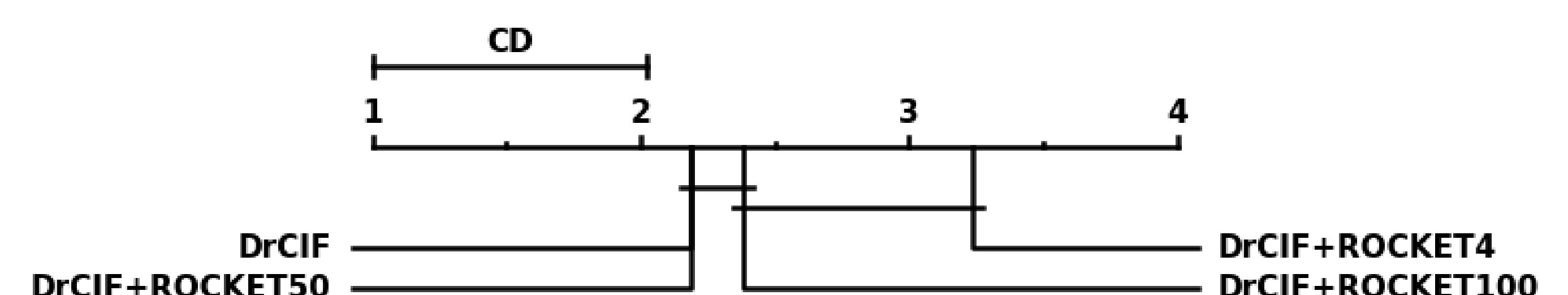


Figure 3: Critical Difference Diagram for average accuracy.

The diagram shows that DrRocket with 50 kernels achieved higher average performance than the other variants and is statistically similar to the original DrCIF, but with much greater efficiency.

Conclusion

The proposed modification, replacing the DrCIF features with those from ROCKET, proved to be viable. DrRocket with 50 kernels:

- Significantly reduced execution time;
- Delivered consistent results;
- Showed a balance between cost and performance, being more promising than the 4- and 100-kernel variants.

DrRocket with 50 kernels proved to be statistically equivalent to DrCIF, standing out for its efficiency and lower execution cost, which reinforces its potential for real-life applications, where speed and accuracy are essential.

References

- [1] Alejandro Pasos Ruiz, Michael Flynn, James Large, Matthew Middlehurst, and Anthony Bagnall. The great multivariate time series classification bake off: a review and experimental evaluation of recent algorithmic advances. *Data Mining and Knowledge Discovery*, pages 1–49, 2020.
- [2] M. Middlehurst, J. Large, M. Flynn, J. Lines, A. Bostrom, and A. Bagnall. Hivecote 2.0: A new meta ensemble for time series classification. *Machine Learning*, 110(11):3211–3243, 2021.
- [3] A. Dempster, F. Petitjean, and G. I. Webb. Rocket: Exceptionally fast and accurate time series classification using random convolutional kernels. *Data Mining and Knowledge Discovery*, 34(5):1454–1495, 2020.